EE3032 Introduction to VLSI Design—Homework 1

1. **Static CMOS Logic** (4 points)
   Realize each of the following Boolean functions using static CMOS logic: (a) $f(w,x,y,z) = \overline{x+y+w\cdot y\cdot z}$ (b) $f(x,y,z) = \bar{x}\bar{y}+\bar{x}\bar{z}+\bar{y}\bar{z}$. Suppose that only true literals are available. You should show the transistor diagram.

2. **CMOS Design** (3 points)
   Design a CMOS circuit to implement the truth table shown in Table 1. This circuit has 4 data inputs (a_0–a_3), 4 control inputs (c_0–c_3), and 4 data outputs (z_0–z_3).

Table 1: Truth table of a CMOS circuit.

c_3	c_2	c_1	c_0	z_3	z_2	z_1	z_0
1	0	0	0	a_3	a_2	a_1	a_0
0	1	0	0	a_2	a_1	a_0	a_3
0	0	1	0	a_1	a_0	a_3	a_2
0	0	0	1	a_0	a_3	a_2	a_1

3. Consider the circuit shown in Fig. 1. Complete the following questions.
   (a) Determine the logic function F. (1 points)
   (b) Design a CMOS circuit to implement the same logic function. (2 points)

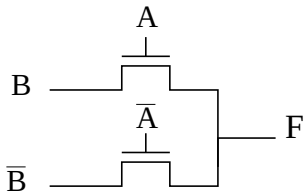

Figure 1: A pass-transistor logic circuit.